

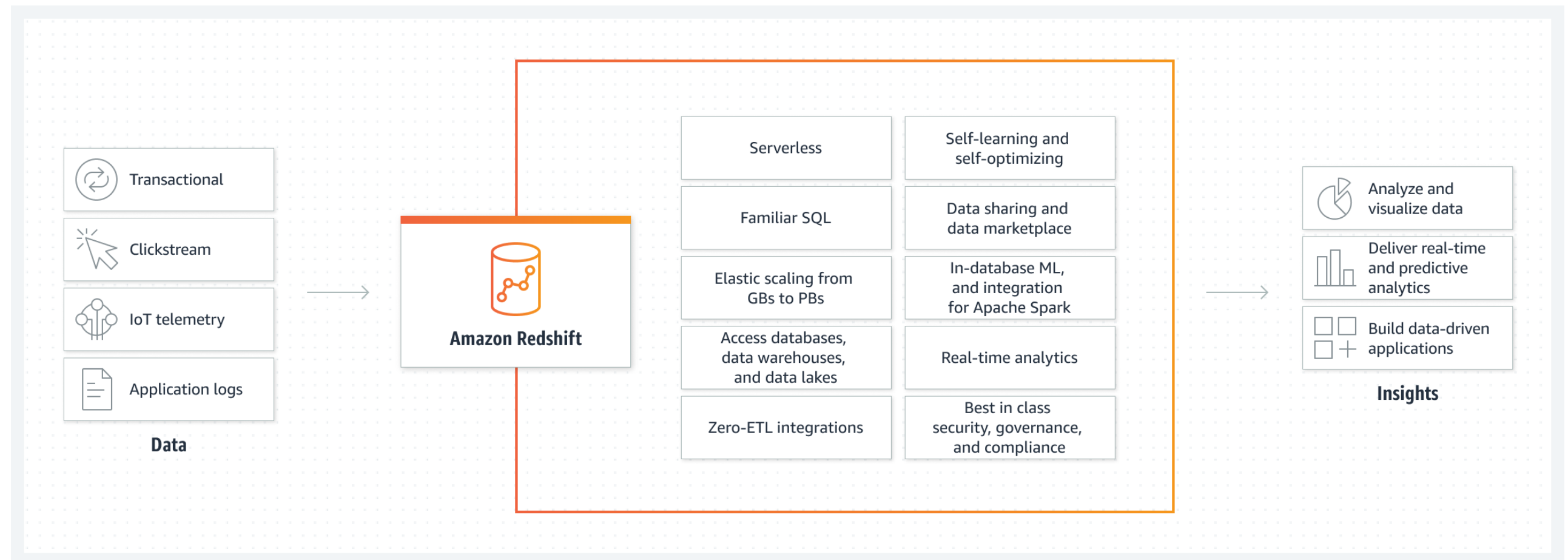
IST — Workshop



IST Workshop

- At the last week of the semester we are going to hold a workshop where groups of students present on a particular topic in data engineering / MLOps.
 - Objective is to go deeper on a concrete cloud service or open source software, based on the theory you saw in class.
- We are going to pretend that you work as a data scientist / data engineer in a Swiss SME and your boss has told you to explore a service / software that appears interesting for the company.
 - Your objective is to find out what is its value for the company, what are its limitations, how much it costs, if there are any alternatives.
 - To be able to go deeper on certain questions, you are going to prepare an **example use case** that demonstrates how the service / software is used.
 - Your boss wants a **critical evaluation** of the service / software. Don't fall for content produced by a marketing department. Distinguish important features from unimportant features. Analyse the service / software as an engineer.
 - At the last week of the semester you are going to **present** your findings in a 15-minute presentation. This presentation includes a **demo** of the use case.

AWS Redshift cloud data warehouse



- Redshift is a data warehouse cloud service
- Based on PostgreSQL
- Can be deployed as dedicated cluster or as serverless service
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other AWS services
- Scales to Petabytes of data

AWS Redshift cloud data warehouse

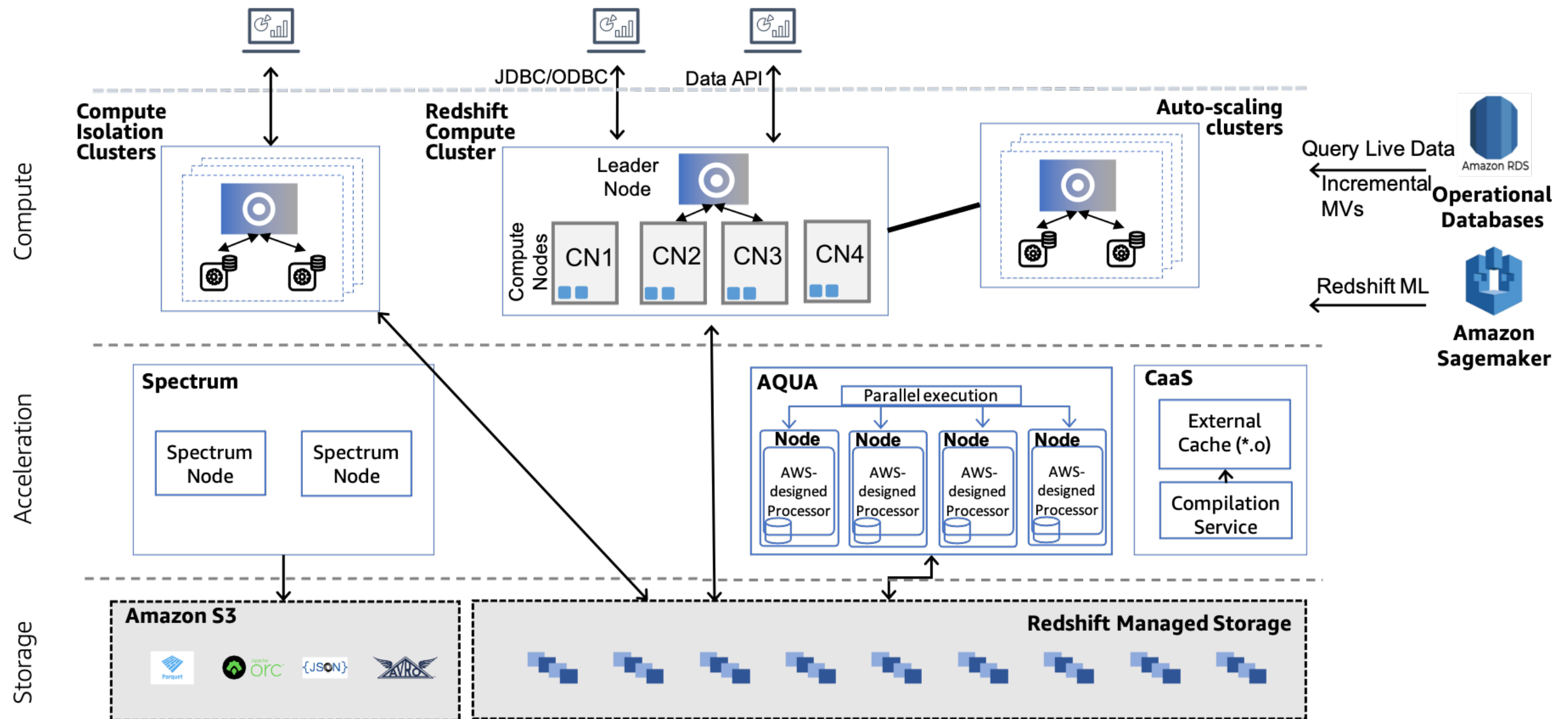


Figure 1: Amazon Redshift Architecture

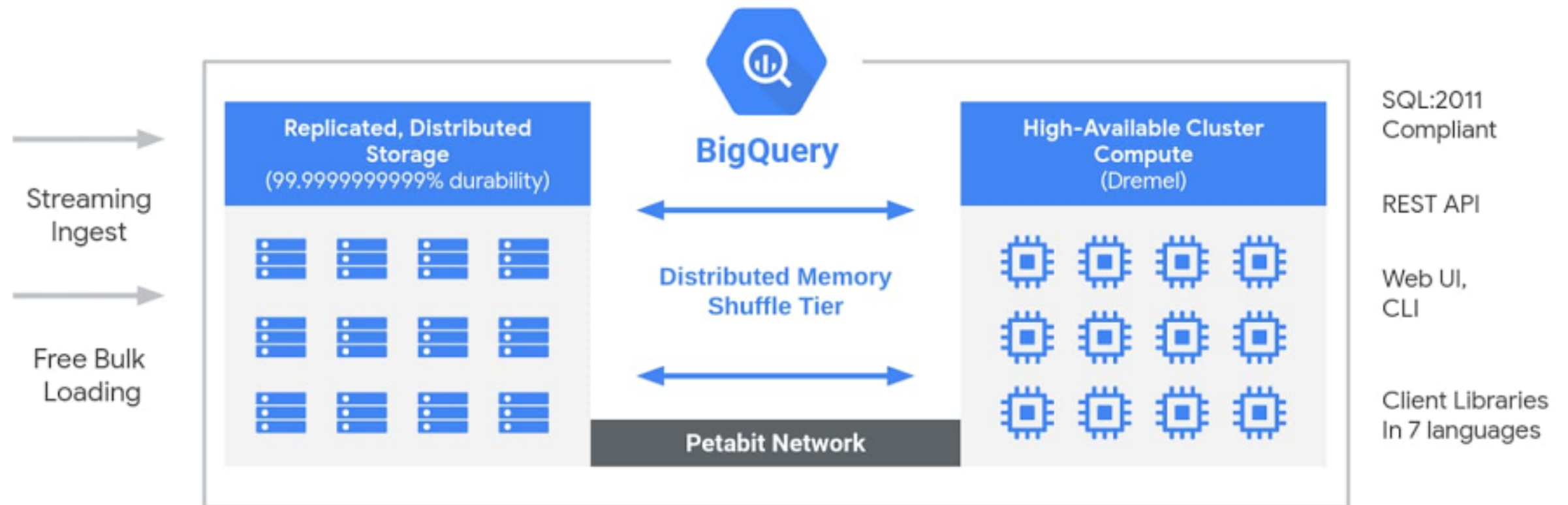
Google BigQuery cloud data warehouse

The screenshot shows the Google Cloud BigQuery interface. The top navigation bar includes the Google Cloud logo, a dropdown menu for 'dbt-graf', a search bar, and user profile icons. The left sidebar contains navigation links for BigQuery, Analysis, Data transfers, Scheduled queries, Analytics Hub, Dataform, Migration, SQL translation, Administration, and Monitoring. The main Explorer panel shows a search bar and a list of resources. The 'bigquery-public-data' dataset is expanded, showing a list of datasets including 'america_health_rankings', 'austin_311', 'austin_bikeshare', 'austin_crime', 'austin_incidents', 'austin_waste', 'baseball', and 'bitcoin_blockchain'. The 'ghcnm_tmax' dataset is selected, and its preview is shown in the right panel. The preview table has columns: Row, id, year, element, value1, dmflag1, and qcflag1. The data shows 12 rows of TMAX values for various years and locations.

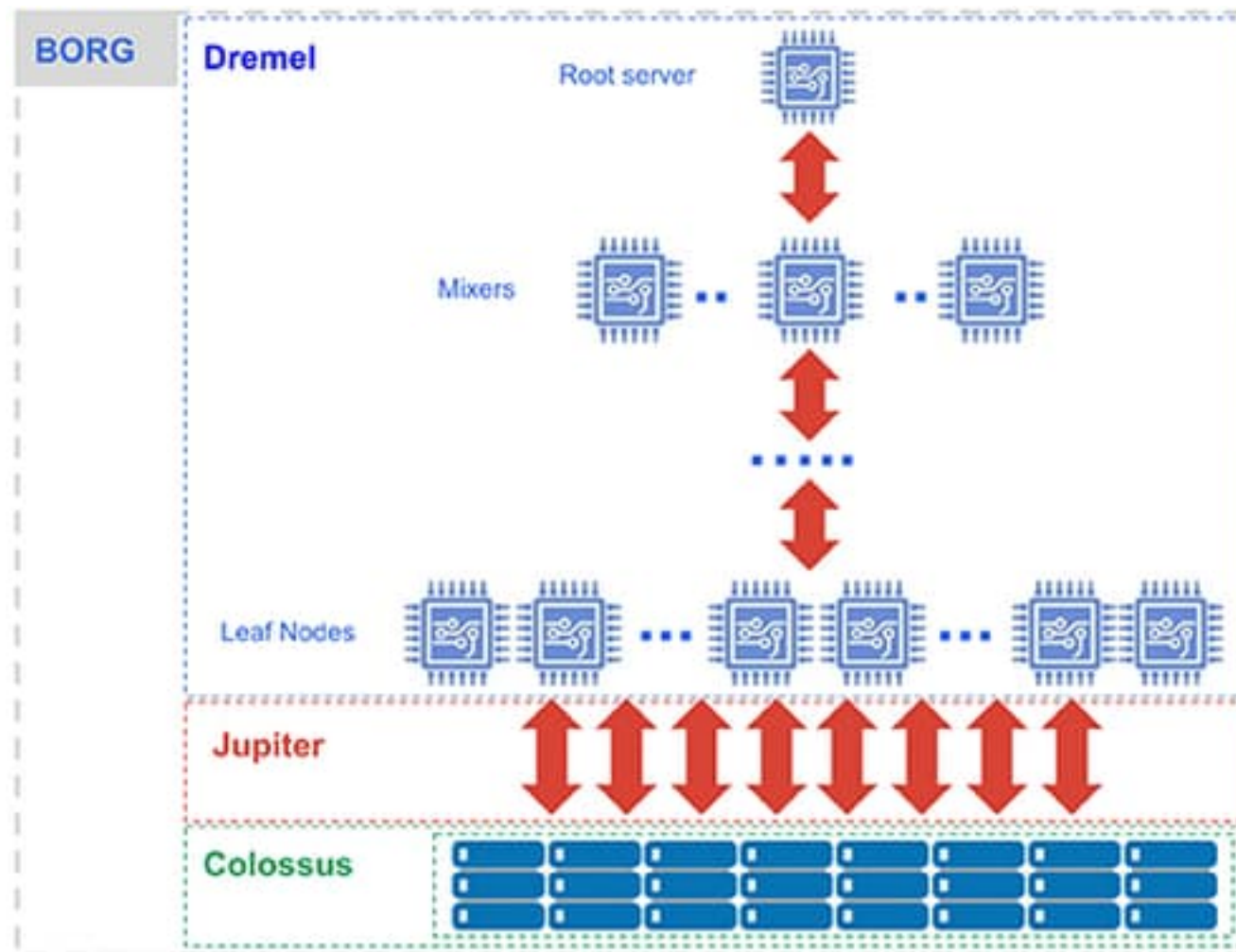
Row	id	year	element	value1	dmflag1	qcflag1
1	10160590000	2012	TMAX	1550	null	null
2	10565516000	1997	TMAX	-9999	null	null
3	11264400000	2004	TMAX	3050	null	null
4	11763450000	2016	TMAX	-9999	null	null
5	12462010000	2000	TMAX	1630	null	null
6	12462012000	1939	TMAX	1920	null	null
7	12761272000	2012	TMAX	-9999	null	X
8	13361090000	1998	TMAX	2930	null	null
9	13465010000	1939	TMAX	3867	null	null
10	13761695000	2016	TMAX	3710	null	null
11	13761698000	1999	TMAX	-9999	null	null
12	14862641000	2019	TMAX	-9999	null	X

- BigQuery is a data warehouse cloud service
- Based on proprietary Google technologies: Colossus, Jupiter, Dremel and Borg
- Serverless
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other Google Cloud services
- Scales to Petabytes of data

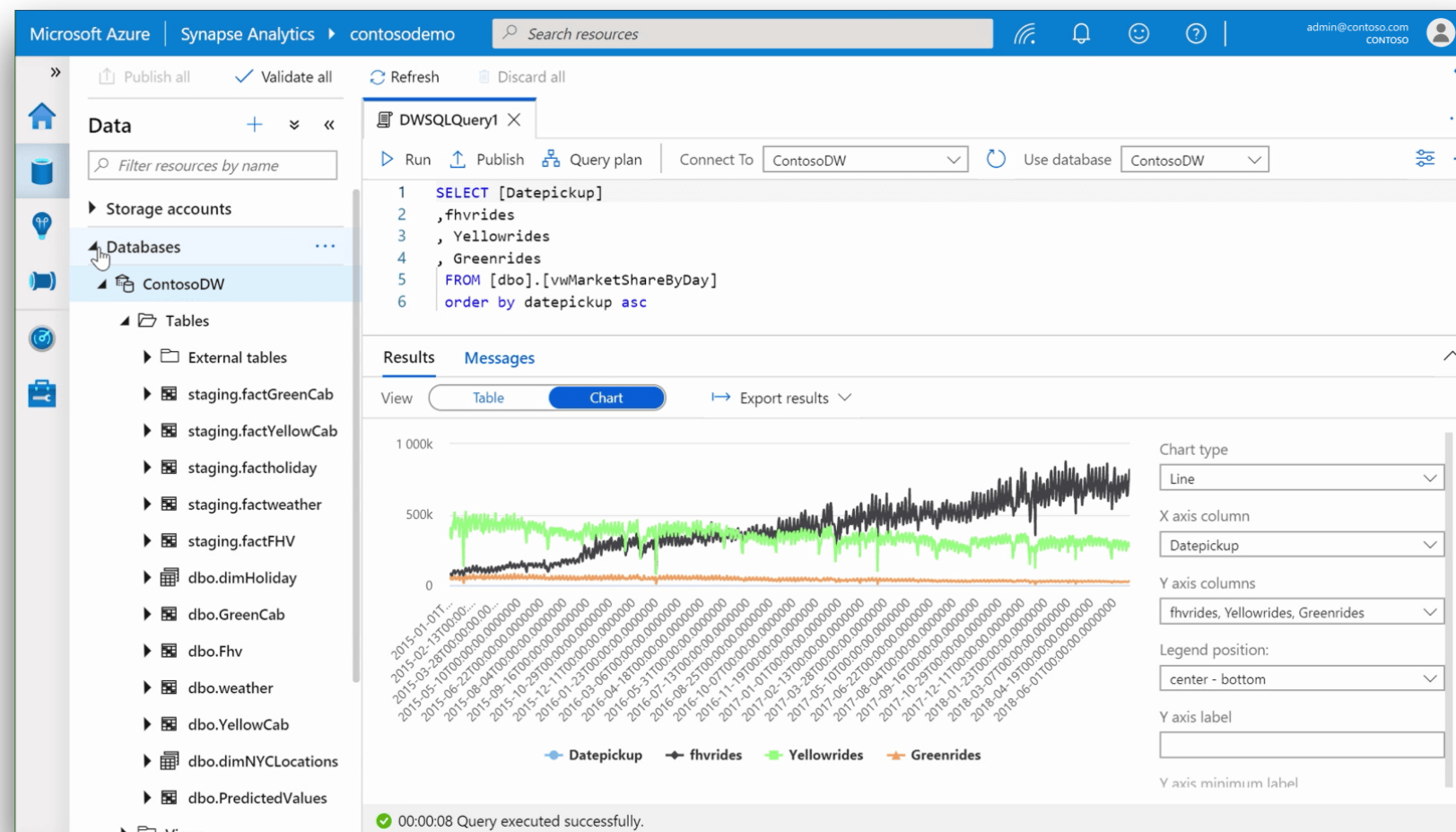
Google BigQuery cloud data warehouse



Google BigQuery cloud data warehouse

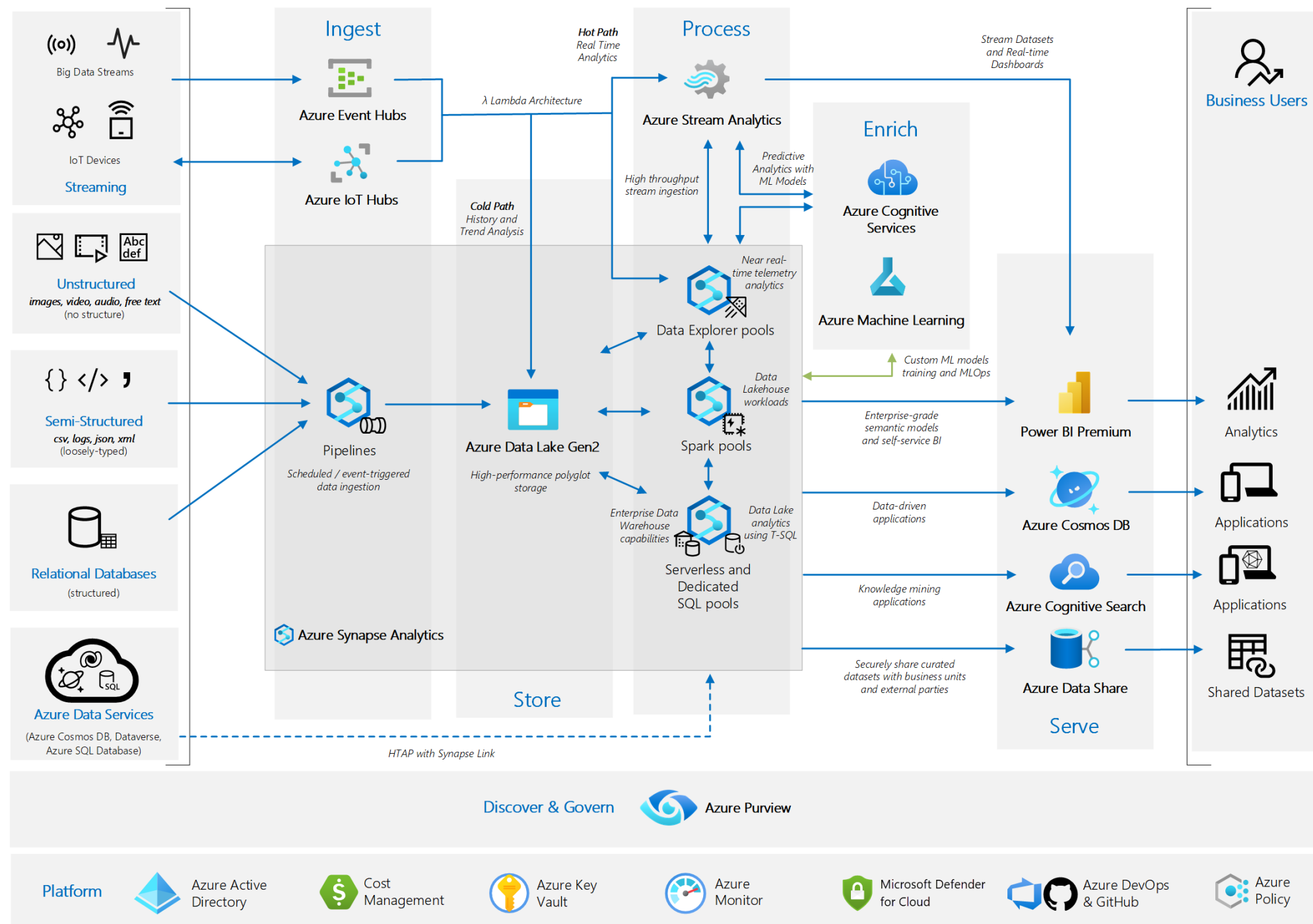


Azure Synapse Analytics cloud data warehouse

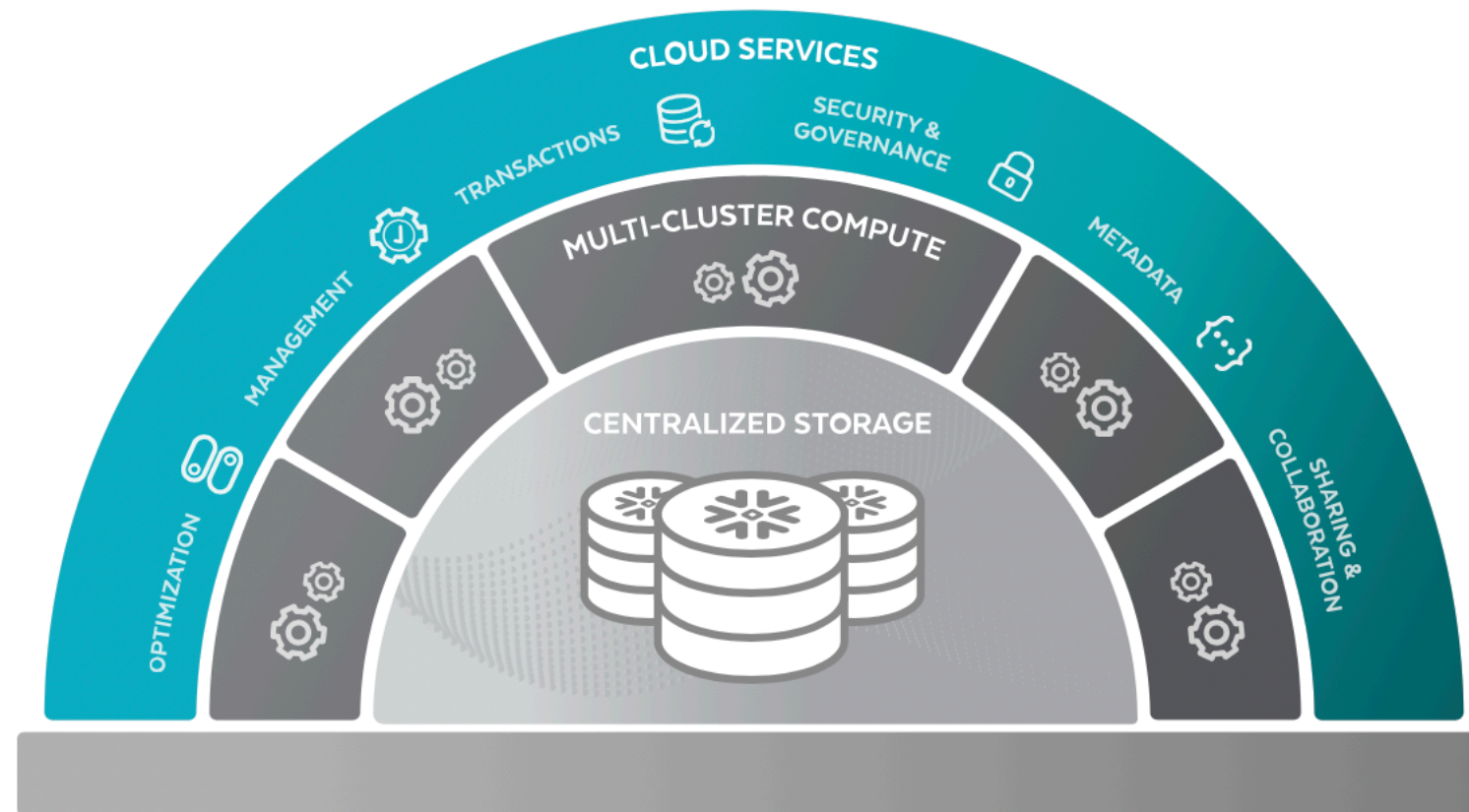


- Synapse Analytics is a data warehouse cloud service
- Can be deployed as dedicated cluster or as serverless service
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other Azure services
- Scales to Petabytes of data

Azure Synapse Analytics cloud data warehouse

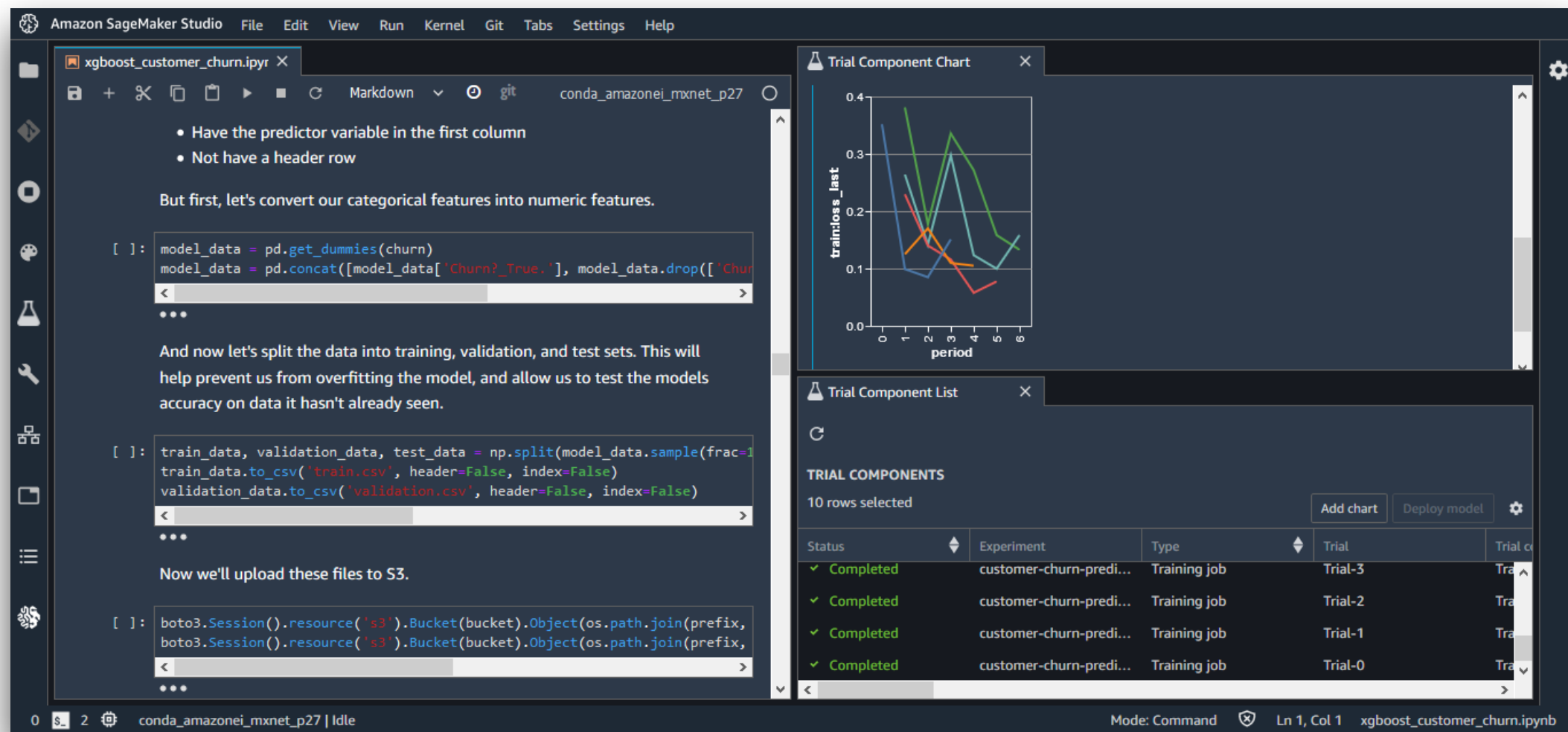


Snowflake cloud data warehouse



- Snowflake is a data warehouse cloud service
- Massively Parallel Processing (MPP) SQL engine
- Scales to Petabytes of data

Amazon SageMaker machine learning platform



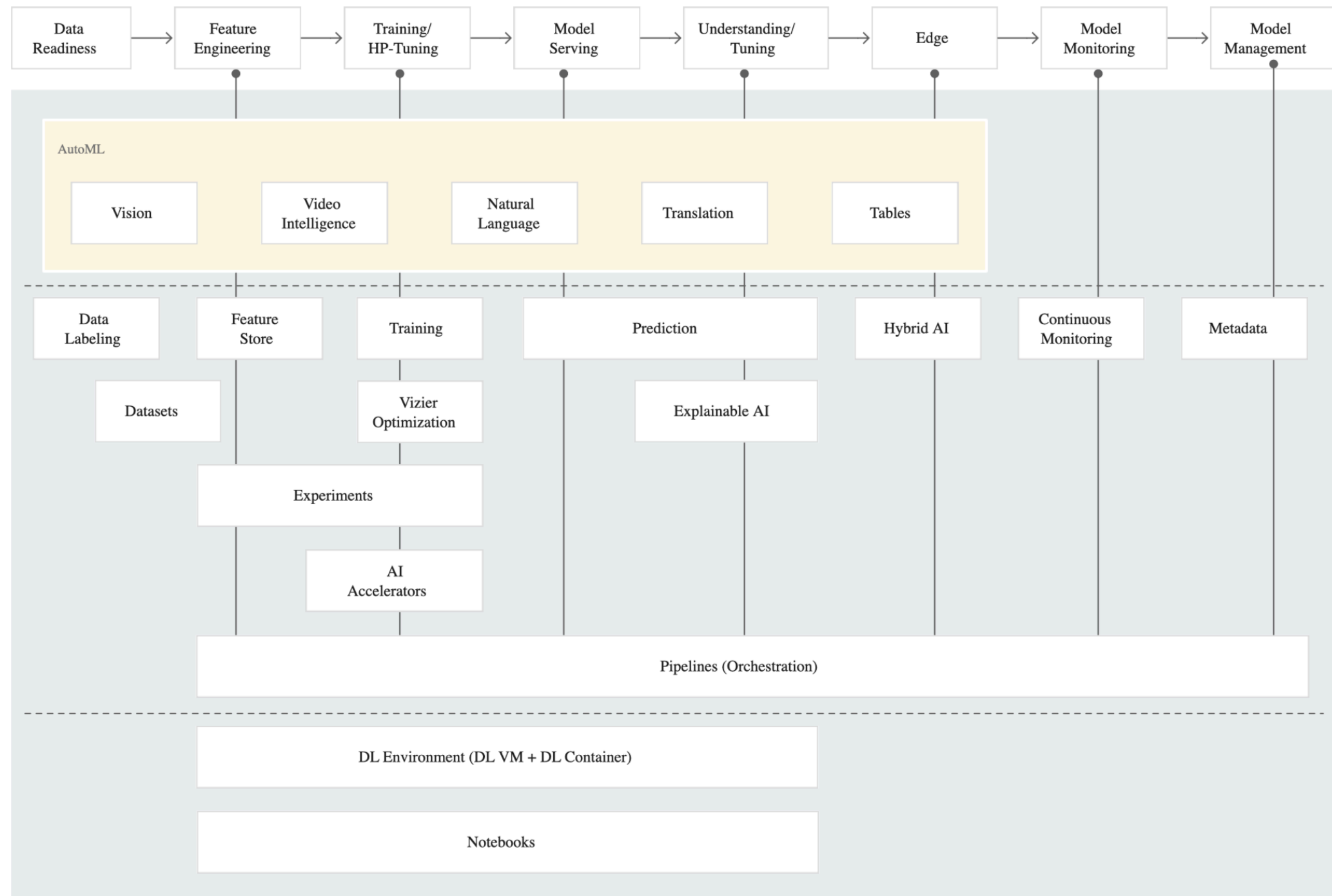
- SageMaker is a machine learning platform in the cloud
- Prepare data, build, train, and deploy machine learning models

Amazon SageMaker machine learning platform

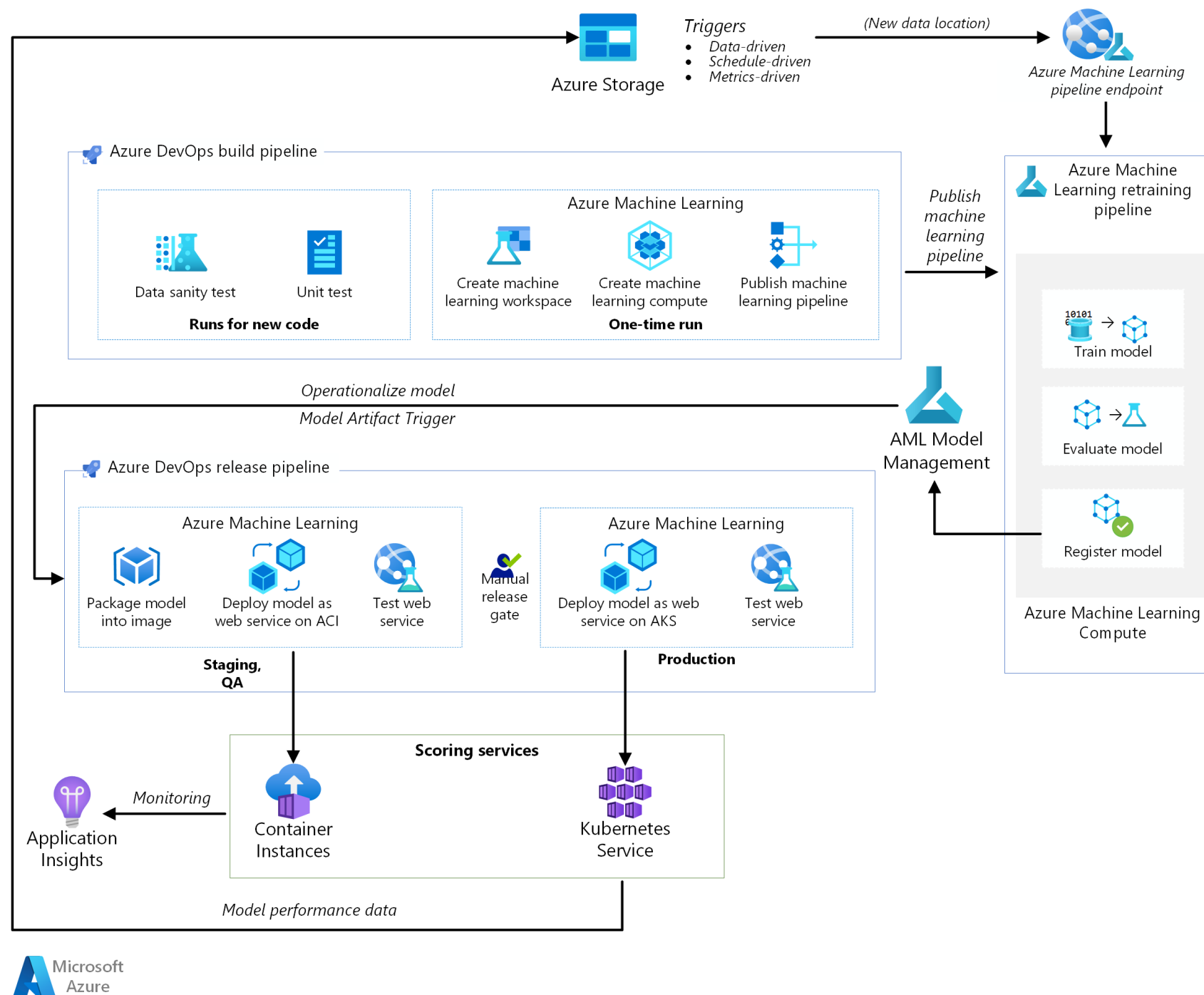
Features

- SageMaker **Notebooks** with Python, Spark and R kernels
- SageMaker **Clarify** detects statistical bias across the entire ML workflow and includes tools to help explain ML models
- SageMaker **Studio** provides a single UI for all ML development steps. You can quickly upload data, create new notebooks, train and tune models, move back and forth between steps to adjust experiments, compare results, and deploy models to production all in one place
- SageMaker **Model Cards** capture model metadata, which can be shared between developers
- SageMaker **Role Manager** provides access control for shared models
- SageMaker **Model Dashboard** gives overview of deployed models
- SageMaker **Autopilot** automatically picks the best algorithm for the data
- SageMaker **JumpStart** provides 150 pre-trained open source models from PyTorch Hub and TensorFlow Hub and allows sharing ML artefacts between developers
- SageMaker **Canvas** is a no-code service with a point-and-click interface to create ML-based predictions
- SageMaker **Pipelines** helps you create fully automated ML workflows. Also provides a model registry.
- SageMaker **DataWrangler** allows you to browse and import data (similar to Glue DataBrew) and create features
- SageMaker **Feature Store** provides a central repository for data features with low latency (milliseconds) reads and writes. Features can be stored, retrieved, discovered, and shared for easy across models and teams.
- SageMaker **Ground Truth** allows you to perform data labelling on images, text files and videos.
- SageMaker **Canvas** is a visual drag-and-drop service that allows business analysts to build ML models and generate accurate predictions without writing any code or requiring ML expertise.
- SageMaker **Experiments** helps you organize and track iterations to ML models. It automatically captures the input parameters, configurations, and results, and stores them as “experiments”.
- SageMaker **Debugger** automatically captures real-time metrics during training, such as confusion matrices and learning gradients, to help improve model accuracy.
- SageMaker **Training Compiler** is a deep learning (DL) compiler that accelerates DL model training by up to 50 percent through graph- and kernel-level optimizations to use GPUs more efficiently.
- For model deployment, SageMakers offers three options
 - Real-time inference
 - Batch transform
 - Asynchronous inference

Google Vertex AI machine learning platform



Azure Machine Learning platform

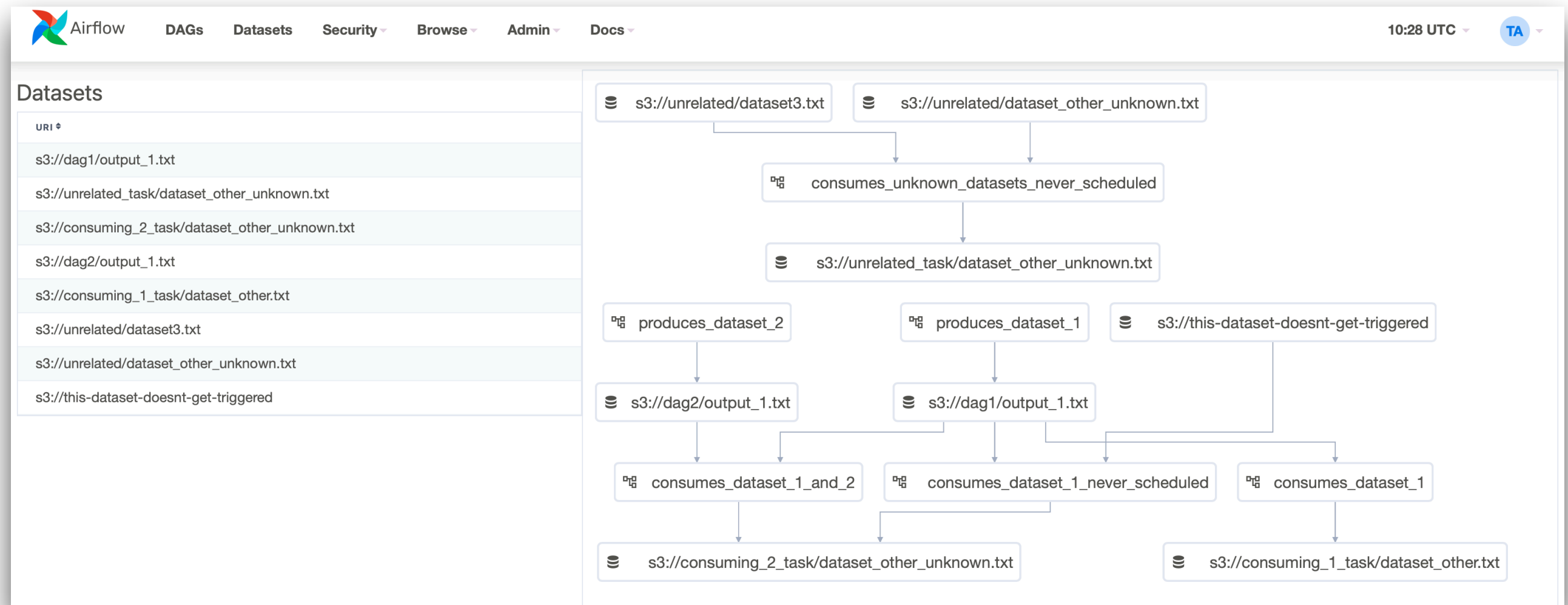


Apache Airflow workflow management software

- Initially developed at Airbnb in 2014 to manage data engineering workflows / pipelines
- 2016 Apache incubator project
- 2019 Apache top-level project
- Written in Python
- Developer defines workflow as directed acyclic graph (DAG)
 - Writes workflow dependencies in Python
 - Writes workflow tasks in Python
- Integration with popular databases, storage services, analytics engines and cloud services via pluggable *operators*.
- Cloud providers offer managed version of Airflow as cloud service
 - Google Cloud Composer
 - AWS Managed Workflows for Apache Airflow



Apache Airflow workflow management software



Apache Airflow workflow management software

The screenshot displays the Apache Airflow web interface. At the top, the navigation bar includes the Airflow logo, links to DAGs, Security, Browse, Admin, and Docs, the current time (10:46 PDT (-07:00)), and a user profile (FB). The main header shows the DAG name 'example_bash_operator' with a 'success' status and a 'schedule: 0 0 * * *' cron expression.

Below the header, there are several tabs for viewing the DAG: Tree, Graph (selected), Calendar, Task Duration, Task Tries, Landing Times, Gantt, Details, and Code. To the right of these tabs are buttons for 'Run', 'Refresh', and 'Delete'.

The 'Graph' tab displays a DAG visualization. The DAG consists of several tasks: 'runme_0', 'also_run_this', 'runme_1', 'run_after_loop', 'runme_2', 'this_will_skip', and 'run_this_last'. The tasks are connected by arrows indicating the flow of the workflow. The 'runme_0', 'runme_1', and 'runme_2' tasks are green, indicating they are successful. The 'also_run_this' task is yellow, indicating it is in a 'queued' state. The 'run_after_loop' task is green, indicating it is successful. The 'this_will_skip' task is pink, indicating it is in a 'skipped' state. The 'run_this_last' task is green, indicating it is successful.

At the bottom of the DAG visualization, there is an 'Auto-refresh' toggle switch and a 'Refresh' button.

Apache Airflow workflow management software


Airflow

[DAGs](#)
[Security](#)
[Browse](#)
[Admin](#)
[Docs](#)

21:11 UTC

RH

DAGs

All 26

Active 10

Paused 16

Filter DAGs by tag

Search DAGs

DAG	Owner	Runs	Schedule	Last Run	Recent Tasks	Actions	Links
☒ example_bash_operator example example2	airflow	2	00***	2020-10-26, 21:08:11	6	▶ ↺ 🗑️	...
☒ example_branch_dop_operator_v3 example	airflow		*/* ****			▶ ↺ 🗑️	...
☐ example_branch_operator example example2	airflow	1	@daily	2020-10-23, 14:09:17	11	▶ ↺ 🗑️	...
☒ example_complex example example2 example3	airflow	1 1	None	2020-10-26, 21:08:04	37 37	▶ ↺ 🗑️	...
☒ example_external_task_marker_child	airflow	1	None	2020-10-26, 21:07:33	2	▶ ↺ 🗑️	...
☒ example_external_task_marker_parent	airflow	1	None	2020-10-26, 21:08:34	1	▶ ↺ 🗑️	...
☒ example_kubernetes_executor example example2	airflow		None			▶ ↺ 🗑️	...
☒ example_kubernetes_executor_config example3	airflow	1	None	2020-10-26, 21:07:40	5	▶ ↺ 🗑️	...
☒ example_nested_branch_dag example	airflow	1	@daily	2020-10-26, 21:07:37	9	▶ ↺ 🗑️	...
☐ example_passing_params_via_test_command example	airflow		*/* ****			▶ ↺ 🗑️	...

Apache Airflow workflow management software

queued

running

success

failed

up_for_retry

up_for_reschedule

upstream_failed

skipped

scheduled

deferred

no_status

Duration

Apr 06, 20:00

00:02:00

00:01:00

00:00:00

runme_0

runme_1

runme_2

also_run_this

this_will_skip

run_after_loop

run_this_last

DAG

example_bash_operator

Run

2022-04-10, 20:00:00 EDT

DAG Run Details

Graph

Mark Failed

Mark Success

Re-run:

Clear existing tasks

Queue up new tasks

Status: success

Run Id: scheduled__2022-04-10T00:00:00+00:00

Run Type: scheduled

Duration: 00:00:00

Last Scheduling Decision: 2022-04-10, 20:01:12 EDT

Started: 2022-04-10, 20:00:01 EDT

Ended: 2022-04-10, 20:01:12 EDT

Data Interval:

Start: 2022-04-09, 20:00:00 EDT

End: 2022-04-10, 20:00:00 EDT

Auto-refresh

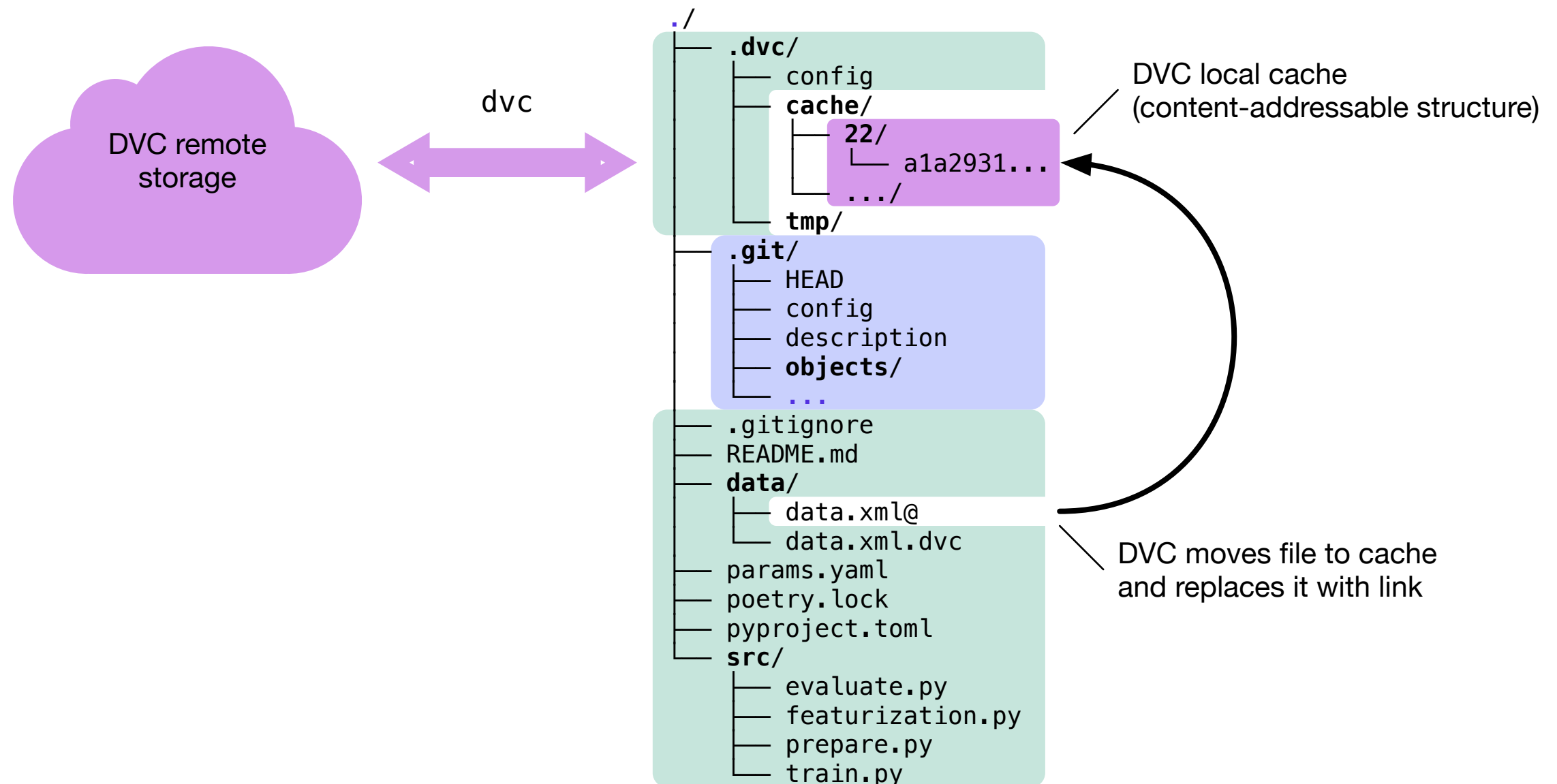
Hide Details Panel

Data Version Control (DVC)

- Open Source project developed by Iterative.ai
 - Initial release 2017-05
 - Written in Python
- Runs on top of any Git repository
 - On the local machine, code and data sets are in the same workspace
 - Data sets are not stored in Git, but only documents pointing to them
 - Data sets are stored in S3 or similar
 - On the local machine, the data is kept in a cache directory (default ``.dvc/cache``). The workspace contains symbolic links to the files in the cache.
- Works as a command-line tool



Data Version Control (DVC)



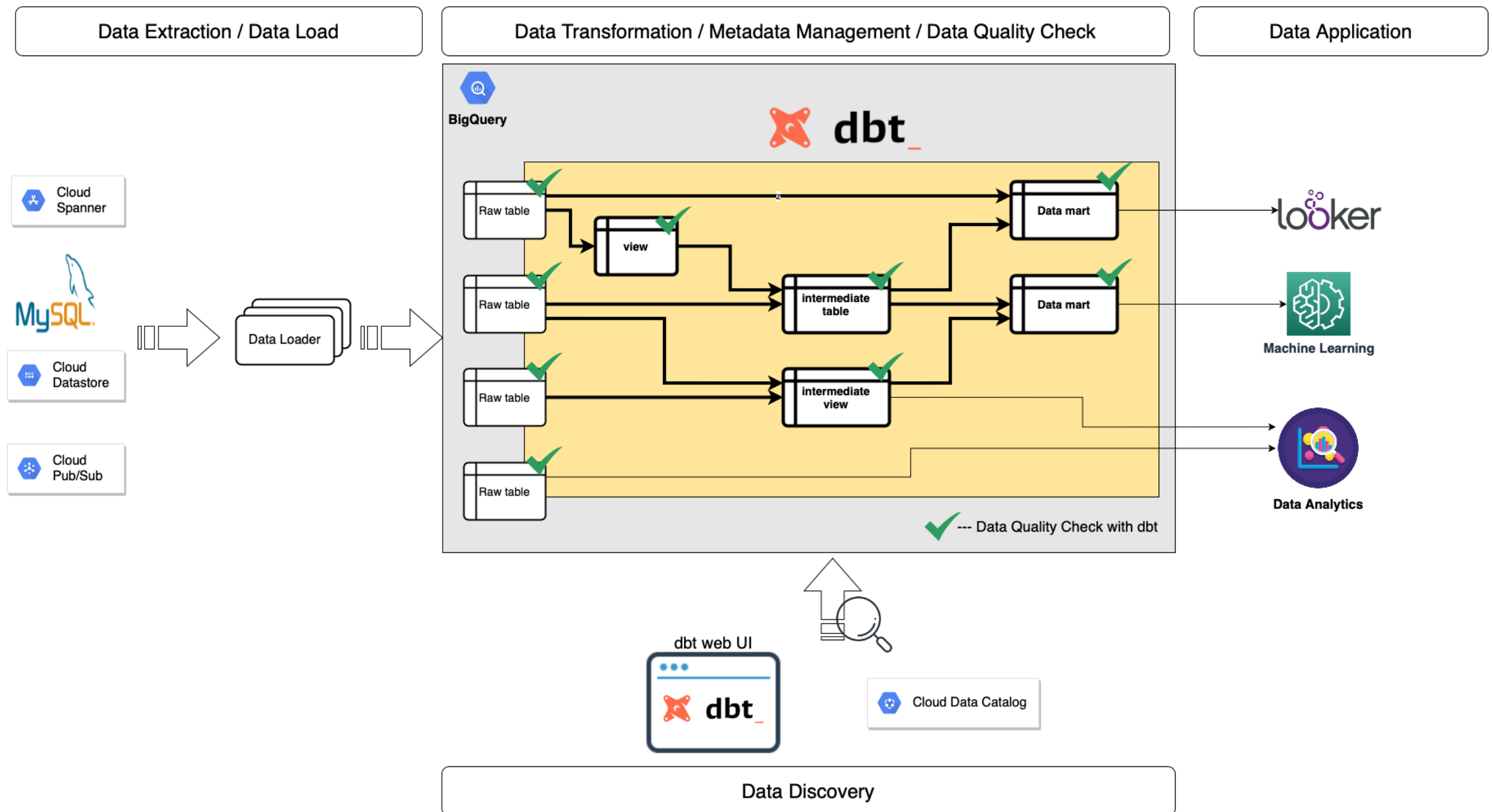
Data Build Tool (dbt)

- Tool for transforming relational data to use it in a data warehouse.
 - Open Source
 - Developed by dbt Labs
- Command-line tool to read source data, transform the data and produce data ready for consumption.
- Transformations are written as SQL SELECT statements.
 - Several transformations can be chained in a pipeline.
 - Several pipelines can share the same transformations.
- To store the data and execute the transformations, dbt connects to a database (local, on premises or in the cloud).
- The SQL code for transformations is stored in a project directory and versioned in Git.



- For “building” the final data, dbt uses the same practices as building an app from source code:
 - dbt automatically manages dependencies between transformations
 - dbt is able to automatically execute data quality tests
 - dbt enables a team to share the build pipeline

Data Build Tool (dbt)



Data Build Tool (dbt)

dbt Docs

getdbt.com/mrr-playbook/#!/model/model.acme.mrr

dbt

Search for models...

Overview

Project Database

Projects

- acme
 - data
 - models
 - customer_churn_month
 - customer_revenue_by_month
 - mrr
 - utils

mrr view

Details Description Columns SQL

Details

TAGS	OWNER	TYPE	PACKAGE	RELATION
untagged	TRANSFORMER	view	acme	analytics.dbt_claire_playbook.mrr

Description

This model represents one record per month, per account (months have been filled in to include any period)

This model classifies each month as one of: new, reactivation, upgrade, downgrade, or churn.

Columns

COLUMN	TYPE	DESCRIPTION
id	TEXT	
date_month	TIMESTAMP_NTZ	
customer_id	NUMBER	

Lineage Graph

customer_revenue_by_month

customer_churn_month

mrr

https://www.getdbt.com/mrr-playbook/#!/model/model.acme.mrr#description